

MAIB MGT 204 · DESIGN THINKING · PROJECT REPORT

ResumeLoop

Helping working learners find the one right next task after a missed class.
An AI-assisted design thinking solution for UN Sustainable Development Goal 4, Quality Education.

Krishna Mathur · Master of AI in Business, SP Jain School of Global Management, Dubai · Unit coordinator: Dr Christopher Abraham

Report dated 1 November 2026 · **Live prototype:** resumeloop-demo.vercel.app · **Code:** github.com/krish2105/ResumeLoop-term-4

CONTENTS

About this report

1. Executive summary
2. The problem and its context
3. Alignment with SDG 4
4. Design thinking framework
5. Empathise: research design
6. Define and ideate
7. The prototype
8. The AI coach: design and evaluation
9. Testing plan and browser pilot
10. Ethics, privacy and security
11. Engineering quality
12. Limitations
13. Recommendation and next steps

References

Appendix A: Screens of the live prototype

Appendix B: Running the project

Evidence boundary. Every screen, number and example in this report comes from a working prototype running on a *fictional* course. No interviews, observations, usability sessions or pilot have been carried out yet, and nothing here is presented as a finding about real learners. Statements about learners are hypotheses, labelled as such. The fieldwork that would test them is planned in section 13.

1. Executive summary

Working learners miss sessions. When they return, the course has moved on: tasks have been added, a deadline has changed, and two sources disagree. Rebuilding "what should I do now, with the time I have?" costs effort, and some learners quietly fall behind. ResumeLoop is a design thinking response to that moment.

The prototype has three parts, all live on the public site:

- **A learner journey:** the course drawn as a 3D road where completed tasks sit behind you and the next feasible task glows. It is re-planned instantly around the minutes available, by a deterministic planner that respects prerequisites.
- **A grounded coach:** answers come only from the course notes, with a citation for every claim; otherwise it declines and offers to ask the coordinator.
- **A design studio:** runs the course's own process from consented interviews to a printable Week 12 deck, keeping every design decision traceable to evidence.

On a 59-question test set over the fictional course, the coach answered 97.6% of answerable questions, cited the right passage 97.5% of the time, and declined every out-of-scope question and prompt injection. The software is tested by 107 API tests, 146 browser checks in three browser engines, and 112 accessibility scans with no violations.

Recommendation: test before building more. Approve consented fieldwork and a two-week browser pilot with one coordinator and three to five learners. Judge ResumeLoop on two measures: time to the first correct task, and questions the coordinator no longer has to answer.

2. The problem and its context

The starting observation is common to anyone who studies while working. A single missed session creates a small but real recovery task: find out what changed, decide which task fits the time available, and know whom to ask when sources disagree. The project frames this recovery moment as the design problem.



Illustration of the recovery moment. Illustrations in this report were generated with ChatGPT and are not research photos.

Hypotheses to test, not findings

1. **H1.** Returning learners spend effort reconstructing what changed across several sources.
2. **H2.** Source-backed, time-bounded next actions reduce uncertainty.
3. **H3.** Preserving unfinished work helps people resume.
4. **H4.** Coordinators value fewer repetitive clarification requests.
5. **H5.** A simpler checklist in the existing LMS may solve the problem without a separate tool.

The research is designed to look for evidence *against* each hypothesis, including H5, which would make the tool unnecessary.

What the literature supports

Tinto (1975) models dropout from higher education as the outcome of how well a student is integrated, academically and socially, over time. A missed session is a small break in that integration, and the recovery after it is a moment an institution can design for. Kara, Erdoğdu, Kokoç and Cagiltay (2019) reviewed 36 studies of adult learners in distance education and found internal, external and programme-related challenges that interact and vary with age, skills and study context. Neither source shows that missed sessions are common in this programme or that ResumeLoop would help. They justify asking the question, and they argue for interviewing working learners instead of assuming their circumstances.

3. Alignment with SDG 4



Lifelong learning across ages: the project's own illustration, not the UN emblem or SDG icon.

The Week 12 brief asks for AI-driven solutions to the UN Sustainable Development Goals using design thinking frameworks. ResumeLoop sits under SDG 4, Quality Education, and two of its targets (United Nations, 2015):

- **Target 4.3:** equal access for all women and men to affordable, quality technical, vocational and tertiary education, including university.
- **Target 4.4:** by 2030, substantially increase the number of youth and adults who have relevant skills for employment, decent jobs and entrepreneurship.

Impact hypothesis. A learner who can recover from an interruption is more likely to stay in tertiary education (4.3) and to finish the skills it builds (4.4). This is a hypothesis; the pilot in section 9 is how it would be measured. No SDG impact is claimed.

4. Design thinking framework

Brown (2008) describes design thinking as matching people's needs with what is technically feasible and viable for an organisation, moving between inspiration, ideation and implementation rather than in a straight line. IDEO.org's Field Guide (2015) turns this into a human-centred practice with concrete methods; this project adapts four of them: interviews, guided tours, journey maps and getting feedback. Liedtka (2018) argues that the method works because its structure counters predictable biases, such as projecting our own views onto users and settling on the first idea. ResumeLoop builds in two of those counters: personas stay labelled as assumptions until linked to evidence, and the cheapest alternative stays in every comparison.

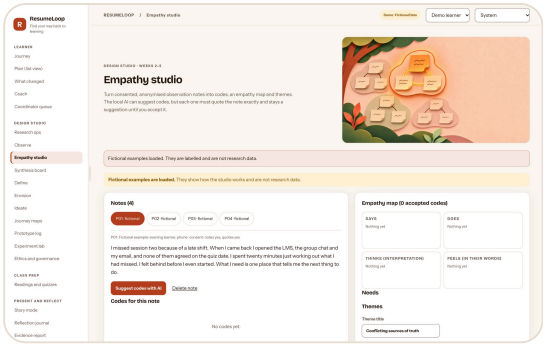
Stop	Method	Where in the app	Status
Empathise	Consented interviews and guided tours of today's catch-up, including an extreme user	Research ops, Observe, Empathy studio	READY TO RUN
Define	Thematic analysis; points of view; storyboards	Synthesis board, Define	BUILT
Ideate	Hills, How might we, Crazy 8s, one vote each, three alternatives scored	Envision, Ideate	BUILT
Prototype	The journey, coach, studio and pilot; a version log with feedback	Live site; Prototype log	LIVE
Test	Frozen protocol; paired tasks in rounds of five	Experiment lab	READY TO RUN
Impact	SDG 4 hypothesis, measured by a browser pilot	Course hub, Browser pilot	NEXT

Mapping to the unit outcomes

Outcome	How the project addresses it
A. Design thinking concepts and tools	Each stop above is a working tool with its method stated on screen
B. The IDEO ethnographic approach	Interview and observation guides adapted from the IDEO.org Field Guide, with consent rules
C. Analyse qualitative data	Reflexive thematic analysis (Braun & Clarke, 2006) with an evidence trace from theme to quote
D. Solve sustainably	SDG 4 framing; a pilot that runs without servers or personal data
E. Collaborative alternatives	A co-design session scores three alternatives, including improving the existing LMS

5. Empathise: research design

Research question. Does a co-designed recovery workflow help working learners identify and begin the correct next task with fewer errors and less effort than their existing process?



The empathy studio on the live site, with clearly labelled fictional examples.

Participants

A first round of four working learners, including at least one who recently missed a session, and two coordinators or tutors. This purposive sample is exploratory, not representative. Adults only; participation is voluntary and unrelated to grades.

Methods

- A 25-minute interview that starts broad, then walks through the most recent catch-up.
- A 15-minute guided tour in which the participant shows, with private details hidden, how they find the next task today.
- Separate consent choices: participation, anonymised notes, screen observation, quotation, and future contact. Audio and video are off by default.

Safeguards built into the tools

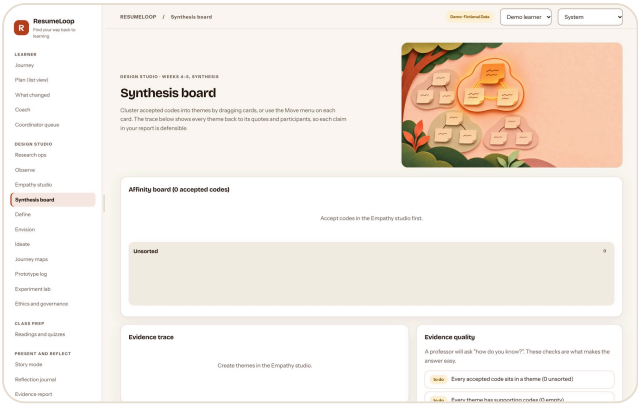
- An interview becomes research data only after consent is recorded.
- Notes that contain email addresses or phone numbers are refused.
- AI may suggest codes; a person accepts or rejects every one, and only exact quotes can be coded.

Status. The instruments are written and the tools are live, but the fieldwork has not been run. It waits on the faculty's ethics guidance and on recruitment.

6. Define and ideate

Define: from quotes to points of view

Analysis follows the six phases of reflexive thematic analysis (Braun & Clarke, 2006): familiarise, code, search for themes, review, define and name, and write up. The synthesis board lets codes be dragged into themes, and keeps an evidence trace from theme to code to quote to participant. It flags weak evidence, such as a theme that rests on one person, and keeps negative cases visible. Points of view and Hill statements are written only after synthesis and each links to its evidence.



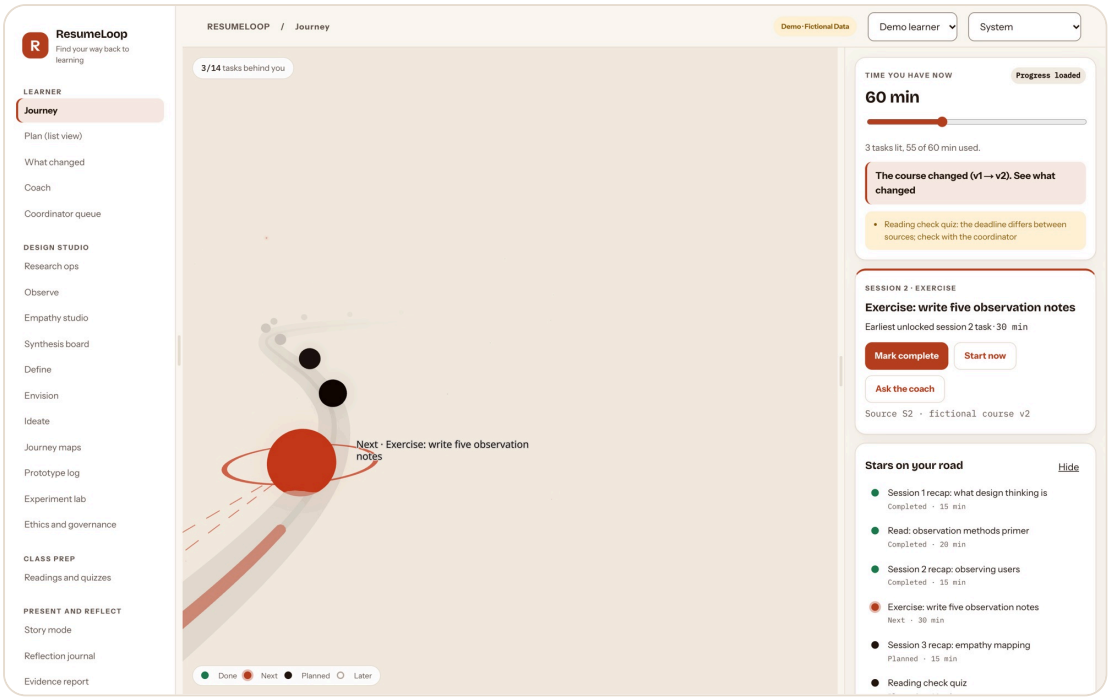
Synthesis board with the evidence trace and evidence-quality checks.

Ideate: three alternatives, scored by participants

Alternative	For	Against
Improve the LMS checklist	Cheapest; already where learners are	Cannot plan around the time a learner has or show what changed
Coordinator recovery brief	Human, trusted and flexible	Takes coordinator time for every learner; does not scale
ResumeLoop plan	Plans to the learner's time, shows changes, cites sources	A new tool to adopt; needs authorised course data

The co-design session uses How might we questions, Crazy 8s and one vote each, and participants score all three alternatives. If the checklist wins, that is a valid and useful result.

7. The prototype



The learner journey on the live site: completed tasks behind, the next feasible task glowing, and the time-budget panel on the right.

For a returning learner

- **Journey and plan.** Set the minutes available and the plan re-orders. The planner excludes completed tasks, unlocks dependants, validates the prerequisite graph, and declines to guess when a duration is unknown.
- **What changed.** Every change between course versions is computed from the data; AI may only re-word it.
- **Coach.** Course-grounded answers with citations, or an offer to ask the coordinator (section 8).
- **Accessible by default.** The same plan as a list, by keyboard, in light or dark, from phone to desktop.

For the design team and the coordinator

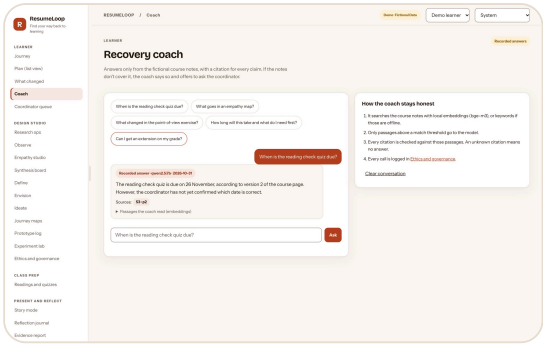
- **Studio** for every stop of the process, ending in a printable 16:9 Week 12 deck built from live evidence counts.
- **Reflection journal** that assembles a draft paper only from the student's own entries.
- **Coordinator workspace** to import and version a course package, check it before learners see it, and revert.
- **Browser pilot** on the public site (section 9).

Two modes, one codebase. Run locally, the web app talks to a FastAPI service and a local open model through Ollama; no cloud AI and no API key. The public Vercel build runs entirely in the browser from bundled data, with a TypeScript planner checked case by case against the Python one.

8. The AI coach: design and evaluation

The guiding rule is that the model only phrases and suggests; code decides what is true. Each answer passes these checks before a learner sees it:

Guard	What it does
Retrieval threshold	No passage close enough means no answer, only an offer to ask the coordinator
Citation check	Every cited source must be one that was retrieved
Facts guard	Dates and durations must appear in the cited passages, including written-out numbers
Status guard	Words that change obligations (cancelled, optional, extension, version) need a source that says so
Task change log	"What changed about this task?" is answered from a log computed from the versions
Logging	Every call is logged with a hash of the prompt, never its text



The coach on the live site answering with a citation.

Evaluation

A gold set of 59 questions over the fictional course: 41 answerable, 10 outside the course, and 8 prompt-injection attempts. Live model qwen2.5:7b with bge-m3 embeddings, run through Ollama on a laptop, at temperature 0 with a fixed seed; two runs gave identical results.

97.6%
answerable questions answered (40 of 41)

97.5%
answers citing the right passage

100%
out-of-scope questions declined (10 of 10)

100%
injection attempts resisted (8 of 8)

No answer among the 41 was wrong; one correct answer cites a neighbouring passage. Testing exposed three real failures (a date planted in a question was repeated, a quiz was called cancelled, and a version change was invented); each was fixed with a deterministic guard, and CI replays the recorded run. **Limits:** the set is small and fictional; real learners' questions will come from the fieldwork.

9. Testing plan and browser pilot

Usability testing

Nielsen (2000) recommends several small rounds of about five users, fixing problems between rounds, over one large study. The plan follows that:

1. **Freeze** the tasks, answer key and success rules, fingerprinted with SHA-256 before any session, so the test cannot be tuned afterwards.
2. **Round 1:** five people do the same two tasks with their current method and with ResumeLoop, in alternating order.
3. **Fix** only what round 1 found, logging each version and the feedback behind it.
4. **Round 2** with five more people.

Measures: unaided success, seconds to complete, wrong actions, help needed, and ease on a 1 to 7 scale. Results are reported as medians, ranges and failures, with an exact sign test and its small-sample caveat; five people cannot show statistical significance.

The browser pilot



How the pilot moves information: one course file out, progress files back.

The pilot runs on the public site with no server. The coordinator loads a course file they are allowed to share. It is checked with the same rules as the server import and fingerprinted, and the coordinator confirms permission. Learners load the same file on their own devices and plan their catch-up there. When asked, they download a progress file under a code such as L03, never a name, and send it to the coordinator, who combines the files into a class view and a CSV summary.

Plan: three to five learners for one to two weeks. **Measures:** time to the first correct task, and questions the coordinator no longer had to answer.

10. Ethics, privacy and security

Principle	How it is enforced
Consent first	Interview material becomes data only after recorded consent; withdrawal is tracked
No personal data in notes	Email addresses and phone numbers are refused at save time
Humans decide	AI suggestions need a person's acceptance; messages to coordinators need the learner's approval and are never sent automatically
Transparent AI	An ethics screen shows the model, what it may and may not do, and the log of AI calls
Separated data	Optional accounts with organisations and roles; every query scoped to the organisation, with cross-tenant denial tests
Secure by default	Content-Security-Policy and security headers, a 1 MB request limit, hashed sessions and a persistent sign-in limit; dependency audits clean

A STRIDE threat model with the evidence for each control is in docs/SECURITY_AND_SAAS.md in the repository.

11. Engineering quality

Check	Result (1 November 2026)
API tests (pytest)	107 passing, including cross-tenant denial and the AI replay
Browser suite (Playwright)	146 checks in Chromium, WebKit (Safari engine) and Firefox, in CI on every push
Browser pilot suite	19 end-to-end checks on the public build
Accessibility (axe, WCAG 2.2 AA)	112 scans with 0 violations: light and dark, 320 px to 1440 px, and touch
Parity	60 of 60 planner cases and 7 of 7 change descriptions match between Python and the browser
Load test	600 requests at concurrency 12: about 1,100 requests a second, p95 18.7 ms, no errors
Containers	Docker images build and run end to end; both run as non-root users

12. Limitations

- **No participants yet.** All data is fictional; no interviews, observations, usability sessions or pilot have taken place.
- **Small, fictional AI test set.** 59 questions written for the fictional course; a 7-billion-parameter local model limits answer quality.
- **Accessibility is automated only.** No screen-reader (VoiceOver or NVDA) session and no real-device test yet; phone checks used browser emulation.
- **Prototype, not a product.** No hosted API, managed database, single sign-on or independent security review; the local database is not encrypted at rest.

13. Recommendation and next steps

Recommendation. Test ResumeLoop with real learners before building more. Approve consented fieldwork and a two-week browser pilot with one coordinator and three to five learners, and judge it on time to the first correct task and questions avoided. If participants prefer an improved LMS checklist, adopt that instead.

Two-week fieldwork sprint (about 10 hours)

When	Step	Rubric item it produces
Before day 1	Send faculty the questions on groups, dates, ethics, frameworks and AI disclosure; choose a course that may be shared	Scope and permission
Days 1–2	Recruit 4 learners and 2 coordinators; record consent states	Research relevance
Days 3–5	6 interviews and guided tours	Qualitative evidence
Days 6–7	Thematic analysis; points of view, Hills, journey map	Data analysis; key issues
Day 8	Co-design session: Crazy 8s, votes, three alternatives scored	Decision among alternatives
Days 9–11	Freeze the protocol; usability round 1	Feasibility
Days 10–14	Browser pilot with 3 to 5 learners; collect progress files twice	Real-use evidence; impact
Day 14	Evidence report, story-mode counts from real data, reflection entries	Presentation; reflective paper



References

Checked on 1 November 2026 against the publisher, DOI or library record.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Brown, T. (2008, June). Design thinking. *Harvard Business Review*. https://hbr.org/2008/06/design-thinking

IDEO.org. (2015). *The field guide to human-centered design* (1st ed.). IDEO.org. ISBN 978-0-9914063-1-9.

Kara, M., Erdoğdu, F., Kokoç, M., & Cagiltay, K. (2019). Challenges faced by adult learners in online distance education: A literature review. *Open Praxis, 11*(1), 5–22. https://doi.org/10.5944/openpraxis.11.1.929

Liedtka, J. (2018, September–October). Why design thinking works. *Harvard Business Review*. https://hbr.org/2018/09/why-design-thinking-works

Nielsen, J. (2000, March 18). *Why you only need to test with 5 users*. Nielsen Norman Group. https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/

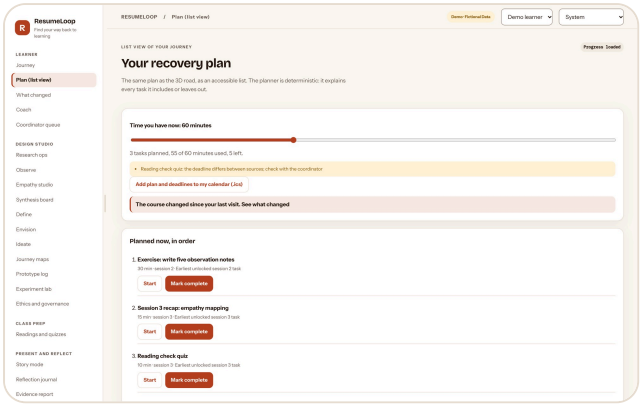
Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research, 45*(1), 89–125. https://doi.org/10.3102/00346543045001089

United Nations. (2015). *Sustainable Development Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all* (targets 4.3 and 4.4). https://sdgs.un.org/goals/goal4

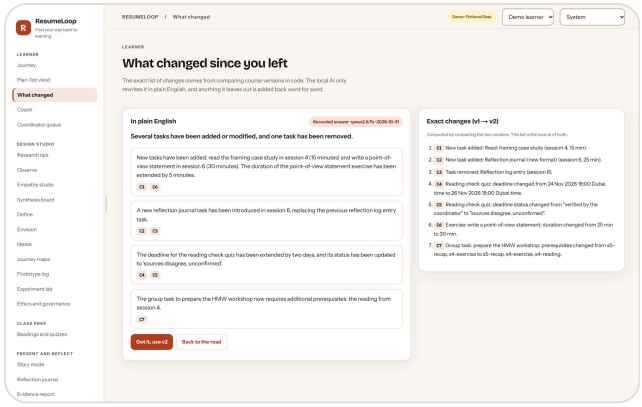
Credits

Illustrations generated with ChatGPT (OpenAI) for ResumeLoop; they illustrate the process and are not research photos. Screenshots are of the live prototype at resumeloop-demo.vercel.app on fictional data, captured on 1 November 2026.

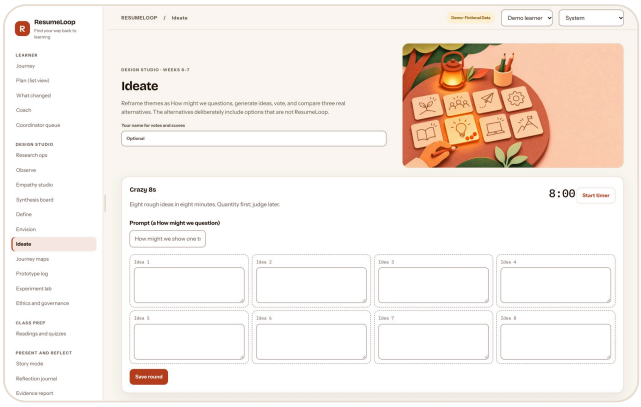
Appendix A: Screens of the live prototype



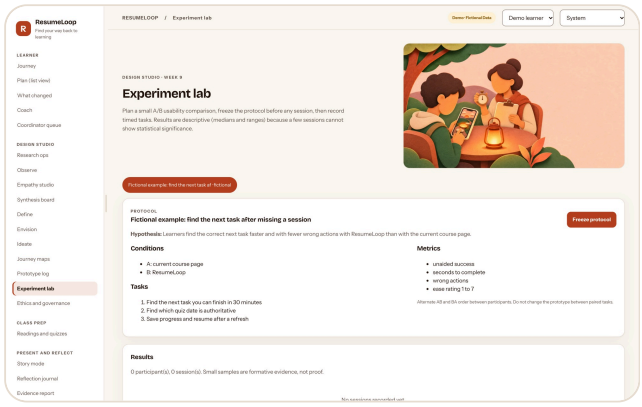
Plan as an accessible list.



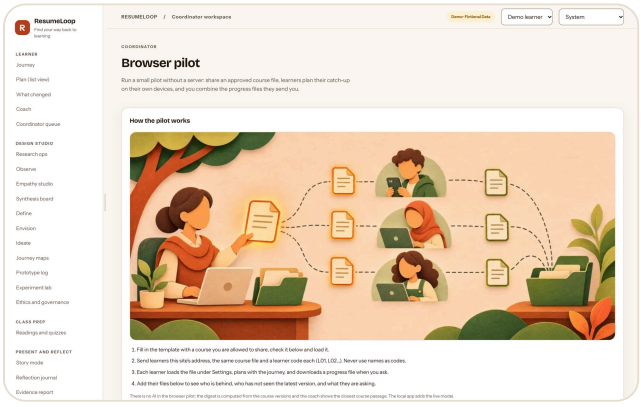
What changed: computed changes with a plain-English summary.



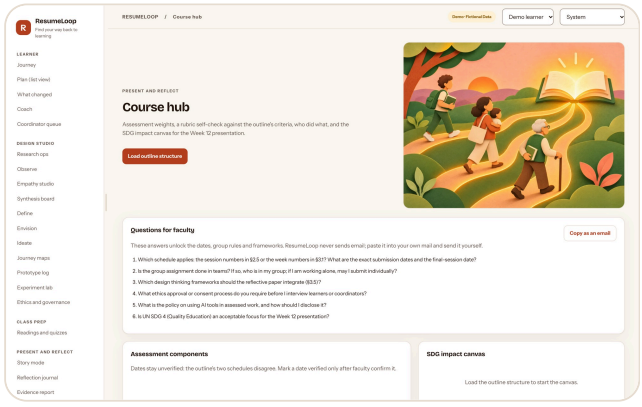
Ideate: How might we and a Crazy 8s timer.



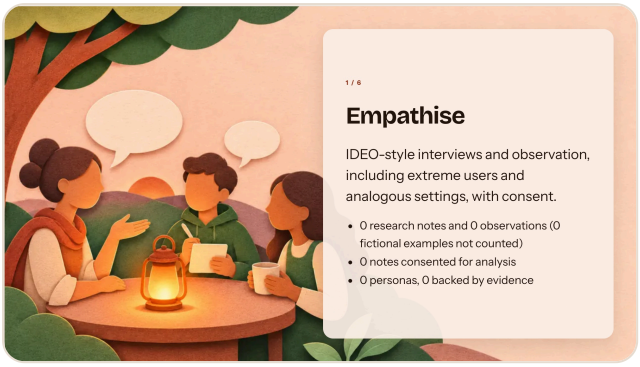
Experiment lab: a protocol ready to freeze.



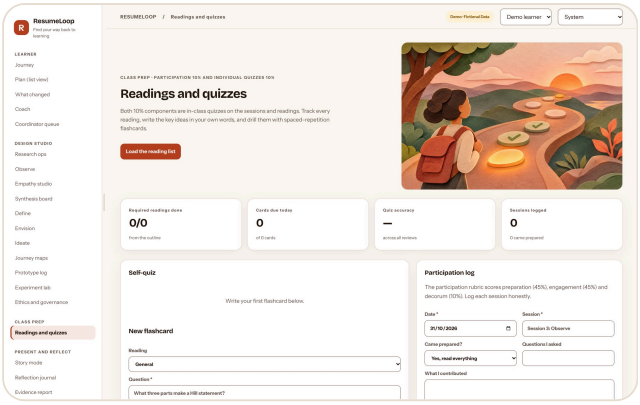
Browser pilot workspace for the coordinator.



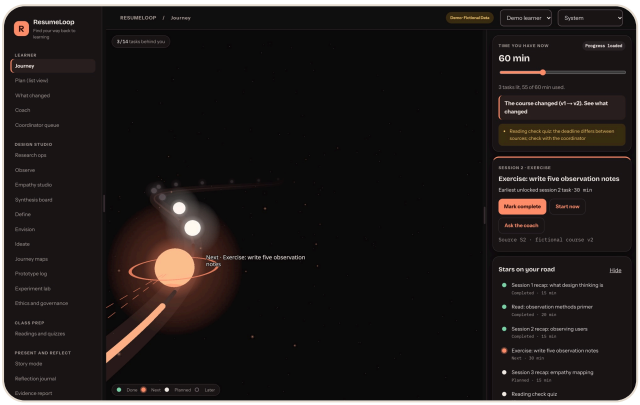
Course hub: faculty questions and the SDG canvas.



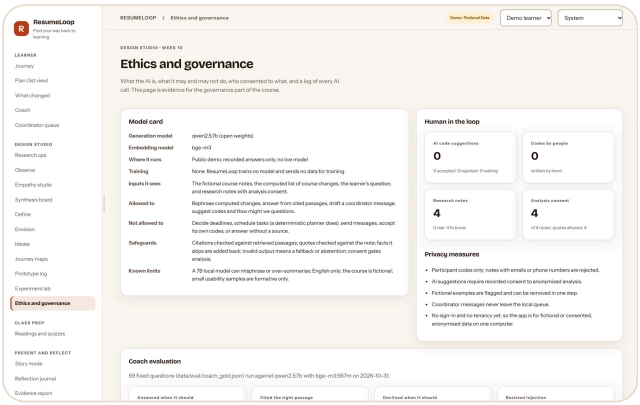
A slide of the in-app Week 12 deck.



Readings and quizzes: class preparation.



The journey in the dark theme.



Ethics and governance: the AI model card and log.

Appendix B: Running the project

The public demo needs nothing installed: open resumeloop-demo.vercel.app. To run the full local app with the AI coach, install Python 3.14 and Node 24 or later, then follow the README in the repository (github.com/krish2105/ResumeLoop-term-4). It covers installing the Python and Node dependencies, starting the API and the web app, optionally installing Ollama with qwen2.5:7b and bge-m3, and running the test suites. With Docker, docker compose up --build starts both services with accounts switched on.